

Perception and adoption of integrated plant disease management by family farmers in Ribeira Valley (São Paulo State, Brazil)

Percepção e adoção de gestão integrada de doenças de plantas por agricultores familiares no Vale do Ribeira (São Paulo, Brasil)

Percepción y adopción de la gestión integrada de las enfermedades de las plantas por los agricultores familiares en el Valle de la Ribeira (Estado de São Paulo, Brasil)

DOI: 10.54033/cadpedv21n3-044

Originals received: 01/16/2024

Acceptance for publication: 02/23/2024

Jonas Akenaton Venturineli Pagassini

Master of Science in Agronomy, Horticulture

Institution: Universidade Estadual Paulista "Júlio de Mesquita Filho", Faculdade de Ciências Agronômicas

Address: Av. Universitária, 3780, Altos do Paraíso, Botucatu – SP, CEP: 18610-034

E-mail: j.pagassini@gmail.com

Luis Carlos Ferreira de Almeida

Doctor of Philosophy in Agronomy, Energy in Agriculture

Institution: Universidade Estadual Paulista "Júlio de Mesquita Filho

Address: Av. Nelson Brihi Badur, 430, Registro – SP, CEP: 11900-000

E-mail: luis.almeida@unesp.br

Maria Candida de Godoy Gasparoto

Doctor of Philosophy in Plant Pathology

Institution: Universidade Estadual Paulista "Júlio de Mesquita Filho"

Address: Av. Nelson Brihi Badur, 430, Registro – SP, CEP: 11900-000

E-mail: maria.candida@unesp.br

ABSTRACT

Family farming, crop diversity and traditional knowledge for plant diseases management characterize Ribeira Valley, São Paulo State, Brazil. Our aim was to diagnose the profile of family farmers from Registro, a municipality of Ribeira Valley, and to determine their perception and adoption of integrated plant disease management (IDM). Semi-structured questionnaire was conducted with farmers and data were analyzed by average, median, mode, Pearson's χ^2 test and Monte Carlo simulation. Men and women are involved in farming, but men are prevalent (68%). There are few young people (18% were up to 30 years

old), reaffirming their exodus from rural activities. These farmers grow and sell vegetable, fruit, medicinal, and ornamental plants, and they exchange information among themselves, with low demand for farm store sellers. Elimination of symptomatic plants is the main control (13% of strategies adopted). Crop residues elimination, crop rotation and resistant plants were also highlighted (12% each one). Most of the respondents did not know IDM concepts, and after the explanation, they agreed that it is possible put into practice. Dialogue between farmers is strategic for them to obtain information, and rural extension is important for the dialogical transmission of knowledge, and for IDM adoption and maintenance in small properties.

Keywords: plant pathology, vegetable diseases, roguing, ethnophytopathology.

RESUMO

A agricultura familiar, a diversidade de culturas e os conhecimentos tradicionais para a gestão de doenças das plantas caracterizam o Vale do Ribeira, no estado de São Paulo, Brasil. Nosso objetivo foi diagnosticar o perfil dos agricultores familiares do Registro, município do Vale do Ribeira, e determinar sua percepção e adoção da gestão integrada de doenças de plantas (MDI). Questionário semiestruturado foi realizado com agricultores e os dados foram analisados por média, mediana, modo, teste χ^2 de Pearson e simulação de Monte Carlo. Homens e mulheres estão envolvidos na agricultura, mas os homens são predominantes (68%). São poucos os jovens (18% tinham até 30 anos) que reafirmam o êxodo rural. Esses agricultores cultivam e vendem plantas vegetais, frutas, medicinais e ornamentais, e trocam informações entre si, com baixa demanda de vendedores de lojas de fazenda. O principal controle é a eliminação de plantas sintomáticas (13% das estratégias adotadas). Também foram destacados a eliminação de resíduos de cultura, rotação de culturas e plantas resistentes (12% cada uma). A maioria dos entrevistados não conhecia os conceitos de IDM e, após a explicação, concordaram que é possível colocá-los em prática. O diálogo entre os agricultores é estratégico para que eles obtenham informações, e a extensão rural é importante para a transmissão dialógica do conhecimento, e para a adoção e manutenção de IDM em pequenas propriedades.

Palavras-chave: fitopatologia, doenças vegetais, malandragem, etnofitopatologia.

RESUMEN

La agricultura familiar, la diversidad de cultivos y el conocimiento tradicional para el manejo de enfermedades de las plantas caracterizan el Valle de Ribeira, Estado de São Paulo, Brasil. Nuestro objetivo fue diagnosticar el perfil de los agricultores familiares de Registro, un municipio del Valle de Ribeira, y determinar su percepción y adopción del manejo integrado de enfermedades vegetales (IDM). Se realizó un cuestionario semiestructurado con los agricultores y se analizaron los datos por promedio, mediana, modo, prueba χ^2 de Pearson y simulación de Monte Carlo. Los hombres y las mujeres participan en la agricultura, pero los hombres son frecuentes (68 %). Hay pocos jóvenes (el 18%

tenía hasta 30 años), lo que reafirma su éxodo de las actividades rurales. Estos agricultores cultivan y venden plantas vegetales, frutales, medicinales y ornamentales, e intercambian información entre ellos, con una baja demanda de vendedores de tiendas agrícolas. El control principal es la eliminación de las plantas sintomáticas (13% de las estrategias adoptadas). También se destacaron la eliminación de residuos de cultivos, la rotación de cultivos y las plantas resistentes (12% cada una). La mayoría de los encuestados no conocían los conceptos de IDM, y después de la explicación, acordaron que es posible ponerlos en práctica. El diálogo entre los agricultores es estratégico para que obtengan información, y la extensión rural es importante para la transmisión dialógica del conocimiento y para la adopción y mantenimiento de la IDM en pequeñas propiedades.

Palabras clave: fitopatología, enfermedades vegetales, roguing, etnofitopatología.

1 INTRODUCTION

More than 60% of the agricultural establishments in Ribeira Valley, São Paulo State, Brazil is composed by small family farmers (IBGE, 2017). In most of the cases, these growers have a high diversity of crops in their establishments and use different intermediate routes linking their production to the final consumer (GOMES; DE ABREU, 2019), such as street fairs (ROMÃO, 2006). According to Romão (2006), Ribeira Valley has low social, economic, and rural development compared to other regions of the State, reflecting the socioeconomic and historical needs of the territory. Almeida *et al.* (2017) highlight several historical events for the development of the region and affirm that any development strategy that does not consider local particularities becomes unsustainable and insufficient for economic and social advances.

For a better understanding of regional demands, it is necessary to identify factors that limit agricultural production to overcome them. Plant diseases is one of many factors that negatively influences the agricultural production, and it is important to avoid this problem or, at least, and more usual, mitigate its effects. The adoption of integrated disease management (IDM) becomes a sustainable ally for growers to mitigate plant disease effects. In the agricultural establishments where IDM practices are adopted, diseases are controlled by integrating a range of methods which consider the economic, social, and

environmental risks. In this context, the idea of eradicating plant diseases is overcome because pathogens are present in the environment and their total elimination, if possible, could result in higher economic, social, and environmental costs than the losses caused by the disease. IDM associates biological, cultural, physical, genetic, and chemical control methods in a holistic way rather than using a single method (AMORIM *et al.*, 2011).

It is possible that the introduction of IDM strategies in the small farms of Ribeira Valley brings benefits to their production chain. Therefore, methodologies that favor the transmission of knowledge through dialogical channels (ZUIN *et al.*, 2011) and that consider the reality of farmers (WÓJCIK *et al.*, 2019; DUFUMIER, 1985) should be preferred. This can result in greater quantity and quality of agricultural products associated with the minimum pesticides use, reducing negative effects on the environment and human health (EL KHOURY; MAKKOUK, 2010).

Some family farmers control plant disease using the same practices as those recommended in the literature, indicating the existence of empirical knowledge applied in their crops. These practices are the result of complex processes in space and time, such as accumulated experience, intrapersonal knowledge, and environmental attributes (WÓJCIK *et al.*, 2019). However, in order to transmit scientific knowledge to rural producers, local demands and specificities should be considered. Almeida (1989) claims that a family farmers profile diagnosis, with information on their agricultural production, constitutes research methodologies whose results help to plan actions in rural areas. In this sense, it is possible to introduce the concepts of the IDM together with an assessment family farmers profile diagnosis to improve agricultural production.

2 THEORETICAL BACKGROUND

2.1 CONCEPTUAL FAMILY FARMING FRAMEWORK AND ITS INSERTION IN RIBEIRA VALLEY (SÃO PAULO STATE, BRAZIL)

Ribeira Valley located in São Paulo State, Brazil, includes 23 municipalities in 1,711,533 hectares (IBGE, 2016). From the 18th century, the period that followed the gold economic cycle, the population of this region started

to specialize in agricultural activities, being, so far, the main economic activity (DIEGUES, 2007). The minimal government support, as well as the region's own social and environmental characteristics, among other factors, diffculted local agricultural development (ROMÃO, 2006; GOMES; DE ABREU, 2019).

In the last Census of Agriculture of the Brazilian Institute of Geography and Statistics (IBGE), there were 5,046 family agricultural establishments in Ribeira Valley, in a total of 8,028 establishments (IBGE, 2017). The census results also show that a large part of agricultural production is sold to distributors or reserved for sales at local fairs and street fairs of rural producers. (ROMÃO, 2006; GOMES; DE ABREU, 2019).

Research about family agricultural production is necessary to know the opportunities and limitations often presented in the farmers reality (MARSDEN *et al.*, 1992). There are many possibilities to categorize social groups in the rural context. Family farming is a diverse and complex field in several ways, such as resources, market access, capital generation and agricultural production (BUAINAIN, 2003). Additionally, in family farming establishment, the family itself is the unit that makes decisions about its own production and activities (SCHNEIDER, 2003).

According to the Brazilian Law 11,326, of July 24th, 2006, the specific parameters to classify family agriculture comprise: limits of up to four fiscal modules (64 hectares in Ribeira Valley, São Paulo State, and the size of the fiscal module unit is changed according to the Brazilian region); agricultural activities mostly carried out by family members; minimum percentage of income generation comes from the agricultural activities carried out in the rural establishment; and management of the rural establishment by the family nucleus (Brazil 2006). Brazilian decree 9,064, of May 31st, 2017, defines family unit for agrarian production as the rural establishment which the exploration of production factors is carried out by family members (who live in or near this place) to their subsistence and/or to society's demands for food and other products/services (BRAZIL, 2017).

This study was based on these legal frameworks to reference the family farmers from Ribeira Valley.

2.2 INTEGRATED DISEASE MANAGEMENT AND ITS IMPORTANCE FOR FAMILY FARMERS IN RIBEIRA VALLEY, SÃO PAULO STATE, BRAZIL

For more than three decades since the 1940s, the objectives of studies on pest control have been basically limited to eradicating the causative agents. This idea was initially reported to pest control because there were low-cost products available, with a wide spectrum of action. The control practices were based on continuous product applications (BERGAMIN FILHO; AMORIM, 1996). However, some negative effects became evident over time, e.g., previously controlled species became recurrent in crops and species of secondary importance started to occur in outbreak levels. Many beneficial biological agents had population reduction and the accumulation of toxic residues in the environment was verified. In consequence, the cost of the products was high and there was no economic viability to continue high frequency of applications (BERGAMIN FILHO; AMORIM, 1996).

At the same time, progress has been made in other control methods as biological control. The idea of maintaining pests at acceptable levels for economic viability was another example of these advances (ZADOKS; SCHEIN, 1979). The control of harmful agents has been added to a broader concept: integrated control, initially defined as a control applied to pests with integrated and combined use of chemical and biological management (STERN *et al.*, 1959). It was not just a simple overlap of two practices, but the integration of all the appropriate practice for each situation (FAO, 1967). Economic and ecological aspects were later added to the concept of integrated control, resulting in integrated pest management. Some years later, the term integrated disease management (IDM) was proposed in the same way, but with another main objective: control of plant pathogenic agents (CHIARAPPA, 1974).

Currently, it is common to plan crops using a farming calendar. The calendar organizes chronologically the rural activities (MONTEIRO, 2009). Using this instrument, the growers know, previously, since the soil preparation period until management after harvest, with indications of fundamental stages such as planting, sanitary treatment, cultural management, among others. Monteiro (2009) also considers the importance of the farming calendar for the production

management plans because it is possible to relate the period of the year with the plant species more appropriate to cultivate, minimizing the risks of losses due to climatic factors. The farming calendar is widely used with different number of activities and operations depending on the specific objectives of each grower. There are discussions to become its use more flexible or even replacing it with disease monitoring, resulting in a rationalization of the use of phytosanitary products, for example (AMORIM *et al.*, 2011).

In Ribeira Valley, São Paulo State, the crop diversity during each year (IEA, 2020) requires different strategies to control plant pathogens. The sequential or simultaneous use of different practices can be a reality in many rural establishments. However, the use of these practices added to the process of growers' raising awareness to IDM importance can result in success: more farmers can adopt these strategies and they can become usual activities in their crops along the years and generations.

The implementation of IDM strategies by family farmers, represents a viable and accessible resource for the management of plant diseases in crops. Additionally, the perception of IDM by the people involved in the agricultural production should be considered to guide extension actions, which may result in the elaboration and adoption of public policies.

In a region where there is a predominance of family farmers, a low adoption of agricultural technologies, and the inevitable incidence of plant pathogens in crops, the following questions arise: (1) The social profile of farmers is related to the actions carried out in their crops, especially regarding to the management of plant diseases? (2) Does the adoption of these actions occur in line with the theoretical descriptions about the IDM?

Our aim was to diagnose the profile of family farmers in Ribeira Valley, São Paulo State, Brazil, and to determine the different knowledge levels, perception and use of IDM strategies in their agricultural properties.

3 MATERIAL AND METHODS

The research consists of the following three steps: sample selection; data collection; data analysis and interpretation (ALMEIDA, 1989). The following sections explain these steps and add descriptions of this work.

3.1 SELECTION AND CHARACTERIZATION OF THE POPULATION SAMPLES

To define the samples to take is not limited to use random units. Researchers should choose the sampling design (random or not) considering which one will collaborate to understand the problem and to answer the research question (CRESWELL, 2010). The possibility of studies with non-random samples denotes the directionality or the intentionality to choose the sampled set. This condition occurs when the parameters for the selection consider the characteristics of the population, the convenience of the study and the knowledge about the studied object by the researcher (MILONE, 2004). Convenience sampling is a non-probability sampling technique where non-random samples are taken simultaneously, unintentional. There is no specific choice of individuals, but the samples are selected from the population only because they are conveniently available to the researcher (LARSON; FARBER, 2010).

The samples of this study were selected by convenience sampling method. The samples were composed of small family farmers. These people represent a heterogeneous group of growers who are responsible for crop diversity in that region, and weekly, they sell their agricultural products in street fairs located in Registro city (24° 29 '15" S, 47° 50 '37" O). Registro is the most important municipality located in Ribeira Valley, São Paulo State, Brazil.

3.2 DETERMINATION OF TECHNIQUES FOR DATA SURVEY

A semi-structured questionnaire was prepared according to the recommendations of Almeida (1989). Questions with an adapted Likert scale were used in the survey (LIKERT, 1932). Data survey took place at street fairs. The brief questionnaire was applied to farmers, individually, in a pre-established time of 5 minutes.

The respondent provides information as: sociodemographic profile, production data, agricultural crops, methods used to control plant diseases, IDM knowledge, and IDM practices. The questionnaire was divided into three sections: (1) Sociodemographic data and production characterization; (2) Agricultural information source, plant disease control methods and frequency and (3) Perception of the IDM (Tables 1, 2 and 3, respectively).

Table 1 – The first section of the questionnaire: Sociodemographic data and production characterisation

Sociodemographic data and production characterisation						
Identification of the farmer and the street fair						
Weekly attendance at street fairs (maximum of 4 days in the week at Registro city)						
1 time	2 times		3 times		4 times	
Age						
Sex and gender						
Education level						
Professional agronomic service in the last 30 days prior to the interview						
Yes			No			
Home institution of the agricultural professional consulted						
CATI ^a	UNESP ^b	APTA ^c	Private company	Self-employed	Agricultural store ^d	Other
Main agricultural species cultivated						

(a) CATI, São Paulo State Coordination of Integral Technical Assistance.

(b) UNESP, São Paulo State University.

(c) APTA, Agribusiness Technology Agency of the São Paulo State.

(d) Centre to sell agricultural goods and services.

Source: the authors, 2024

Table 2 – The second section of the questionnaire: Agricultural information source, plant disease control methods and frequency

Agricultural information source, plant disease control methods and frequency			
Plant disease and its control: Information source and frequency			
	Often or always	Sometimes	Few times or never
Agricultural store seller			
Agricultural Professional			
Internet			
Book			
Other farmers			
Other			
Practises to control plant disease and frequency			
	Often or always	Sometimes	Few times or never
Crop rotation			
Resistant plants			
Elimination of diseased plants			
Seed treatment			
Seedling acquisition			
Replacement of cultivation area			
Replacement of cultivation season			
Fungicides			

Cleaning of agricultural tools			
Elimination of crop residues			

Source: the authors, 2024.

Table 3 – The third section of the questionnaire: Perception of Integrated Disease Management (IDM)

Perception of Integrated Disease Management (IDM)			
Do you know the concept of IDM?			
Yes		No	
Do you think it is possible to adopt IDM?			
I agree	Partially agree	Partially disagree	I disagree
Can you use fungicides as an IDM practise?			
Yes		No	I don't know
Do you have fixed calendar to use fungicides in crop?			
Yes		No	
Would you replace the farming calendar for other models or make it more flexible?			
Yes		Perhaps	No

Source: the authors, 2024.

3.3 DATA SURVEY

Before the survey, visits were made to the street fairs for prior observation of the products offered, in addition to a brief talk with the person who had the role of a leader of the other market farmers. At that moment, the project's aims were presented with an agreement for subsequent visits to apply the questionnaire. The questionnaire was always applied by the same person.

Visits to the market farmers stalls, as well as data survey, took place between June and August 2016 (Figure 1), at the fairs, located in the urban area of Registro city, where they directly sell their agricultural products.

Figure 1 – Visit to the Registro street fairs (São Paulo State, Brazil) in June 2016.



Source: the authors, 2024.

3.4 DATA ANALYSIS AND INTERPRETATION

Sociodemographic data were initially expressed by their descriptive measures: arithmetic average, median and mode.

On the other hand, the relationships between categorical variables will be analyzed by means of Pearson's χ^2 statistical evaluation for contingency tables. Pearson's χ^2 test is based on the differences between observed and expected values, evaluating whether the proportions in each group can be considered similar or not, that is, it verifies the independence between categorical variables (DAWSON; TRAPP, 2003). The p values below 5% were considered significant.

The significant results of the χ^2 test are not able to inform which cells have significant differences between the observed and expected values. In this situation, the analysis of standardized residues will also be applied, considering significant values that are less than -1.96 or greater than 1.96 (SHESKIN, 2003).

In situations where there are cells with expected frequencies less than 2 or 10% of these have frequencies less than 5 and the value of the statistic for the χ^2 test was obtained by Monte Carlo simulation as described by Peat; Barton (2005).

4 RESULTS

4.1 THE FIRST SECTION OF THE QUESTIONNAIRE

At the end of the data survey at the street fairs, 17 family farmers were asked the questions. Their sociodemographic characterization showed a male majority (76%). The arithmetic average, mode and median age of the farmers were 45.12, 37 and 45 years, respectively. In terms of educational level, 47%, 18% and 12% of participants have completed high school, higher education, and middle school, respectively.

Considering the total number of four days of street fairs in a week at Registro city, 41% of participants used to attend three days a week, 18% two days and another 41% used to attend just one day. Only 26% of farmers said they had consulted an agricultural professional to solve a problem or to give some advice about their cultivations during the last 30 days before data survey. These

professionals were consulted only once during this period and, in all these cases, they were self-employed professionals.

The results of the data survey showed the diversity of agricultural species produced by the small farmers located at Ribeira Valley and an offer of 40 products was verified. In general, brassicas (*Brassica* spp.) as *Brassica oleracea* var. *capitata* L., *B. oleracea* var. *botrytis* L. and *B. oleracea* var. *italica* Plenck. were the most mentioned by the growers (produced by 65% of them). Lettuce (*Lactuca sativa* L.), fresh heart of palm (*Bactris gasipaes* Kunth) and eggplant (*Solanum melongena* L.) were cited by 47%, 2a and 18% of participants, respectively. Other species produced in small family farms and cited by questionnaire participants were different species of ornamental, fruit and medicinal plants. Although they were not often mentioned by the rural sellers, these species were frequently offered at the fairs.

4.2 THE SECOND SECTION OF THE QUESTIONNAIRE

In the absolute frequency data of the main sources of technical information, there was a high percentage of farmers who few times or never seek information from agricultural store seller. It was also observed the search for information in books, the internet and consultation with other farmers (Table 4).

Table 4 – Frequency of information search by specialised sources

Information source	Frequency ^(a)		
	Often or always	Sometimes	Few times or never
Agricultural store seller	1	2	14
Agricultural professional	4	8	5
Internet	6	4	7
Book	4	4	9
Other farmers	6	7	1

(a) The frequencies "often or always" and "few times or never" have been associated into a single class, as a way of seeking a distinction amongst classes.

Source: the authors, 2024.

Considering that the χ^2 results for contingency tables alone are not able to indicate the way in which the dependence between the variables occurs, standardized residue analyzes were performed (Table 5).

Table 5 – Analysis of standardised χ^2 residues for all sources consulted

Information source	Standardised residues		
	Often or always	Sometimes	Few times or never
Agricultural store seller	-2,00	ns	3,37
Agricultural professional	ns	ns	ns
Internet	ns	ns	ns
Book	ns	ns	ns
Other farmers	ns	ns	-2,07

Values $> |1.96|$ are significant at 5%; ns = not significant
Source: the authors, 2024.

Negative residual values less than -1.96 indicate that the observed frequencies are statistically lower than expected. Scores higher than 1.96 show that the observed frequencies are significantly higher than expected.

From the results, it is verified that there is no tendency in the search for information with farm store sellers since the values were negative for "Often or always" and positive for "Few times or never". Only one interviewee stated that he always or often seeks information from farm store sellers and, alternatively, 14 stated that they rarely or never consult this source.

To deepen this analysis, a new contingency table was elaborated considering only the two categories that presented significant results (Table 6).

Table 6 – Standardised χ^2 residues analysis for consulted sources

Information sources	Standardised residues		
	Often or always	Sometimes	Few times or never
Farm store seller	-2,45	-2,33	4,17
Other farmers	2,45	2,33	-4,17

Values $> |1.96|$ are significant at 5%.
Source: the authors, 2024.

The positive values for the "Other farmers" x "Often or always" and "Sometimes" relations express that the number of interviewees who prefer other farmers to exchange information is high.

Based on this report on the information sources used by the interviewees, it's possible to discuss decision-making about strategies for the plant disease control (Table 7 and Figure 2).

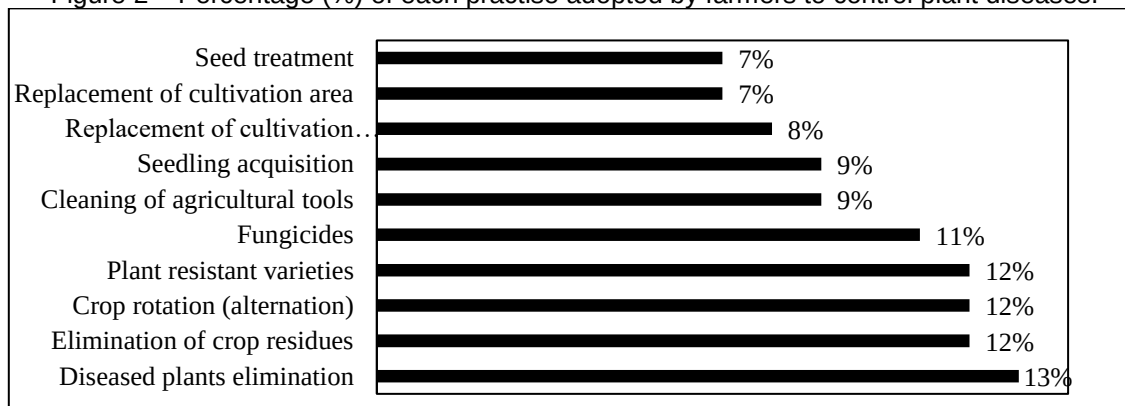
Table 7 – Standardised χ^2 residues analysis for plant disease control practises.

Plant disease control practises	Standardised residues	
	Often or sometimes	Few times or never
Fungicides	ns	ns
Crop rotation / alternation	ns	ns
Resistant plants	ns	ns
Elimination of diseased plants	2,25	-2,25
Seed treatment	-2,35	2,35
Seedling acquisition	ns	ns
Replacement of cultivation area	ns	ns
Replacement of cultivation season	-2,35	2,35
Cleaning of agricultural tools	ns	ns
Elimination of crop residues	ns	ns

Values > | 1.96 | are significant at 5%; ns = not significant
 Source: the authors, 2024.

These results reveal that there is a greater tendency towards Elimination of diseased plants and less tendency towards Seed treatment and Replacement of cultivation season. The other strategies for the control of plant diseases show that the frequency of its use occurs independently because they presented non-significant residue values.

Figure 2 – Percentage (%) of each practise adopted by farmers to control plant diseases.



Source: the authors, 2024.

4.3 THE THIRD SECTION OF THE QUESTIONNAIRE

Only 24% of participants reported formal knowledge about IDM concept. After a brief explanation about this concept, it was asked their opinion about IDM implementation in practice (in the field): 59% answered it is totally possible this implementation, 29% partially possible, 6% partially impossible and 6% totally impossible to execute this in practice.

When asked about the possibility of using fungicides as an IDM strategy, 53% of farmers answered that is allowed to use these inputs and almost 47% were divided exactly between those who did not know and those who thought it was not allowed. It was not possible to identify whether the correct answer was due to a real knowledge of IDM strategies or because it was just an empirically based statement.

Only 12% of farmers stated that they followed a fixed calendar to use fungicides in their crops. The control of diseases by fungicides occurs only when necessary, considering the other management practices that they can use, such as the elimination of diseased plants, as cited previously.

5 DISCUSSION

Many male farmers and little presence of young people are results in agreement with results of Freitas (2015), who observed a majority presence of male farmers (63%) and few young people in Tunas (Ribeira Valley, Paraná State, Brazil) from the analysis of the social profile of a family farmers group registered in the Food Acquisition Program (a Brazilian government program that regulated the food purchase from family farmers).

The age range and balance between men and women in field activities revealed in this study also seem to be related to official data from the Brazilian government, in which data from the last official census revealed that only 19% of rural establishments in Brazil are run by women and most men who run rural establishments are between 45 and 64 years old (IBGE 2017). In view of this, institutional actions and public policies are needed to reduce rural evasion by the youngest and to decrease the gender gap in the conduct of agricultural activities.

The street fairs in Registro are spaces for the diversity of agricultural products, offering fresh plants that, in general, need to be consumed in a few days. There was an expectation that banana (*Musa spp.*) would be one of the agricultural items most produced by those interviewed: the Ribeira Valley has been the largest banana producing region in the state of São Paulo for years (Angelo *et al.*, 2016; IBGE, 2020; IBGE, 2024). Production in this territory influences the national production of bananas (IBGE, 2020 ; IBGE, 2024)..

Despite these data, it was observed a low occurrence of this product in the stalls, probably because banana production has been sold to other markets (wholesale, supermarkets, etc).

The low demand with sellers and the low percentage of farmers who “few times” or “never” seek information from other farmers are data that indicate that there is an exchange of information between farmers as a frequent practice. On the other hand, information from professionals in agricultural sciences, books or websites did not present significant standardized residue values, that is, the frequency of searching for information from these sources occurs independently. The low technical assistance verified in this study agrees with Romão (2006) in the scope of the low regional development, and the distance of rural communities in Ribeira Valley from the state's development processes, reality still occurring in the period of this research.

Empirical knowledge and exchange of knowledge between farmers are decisive factors in the interviewees' agricultural activities. Fernandes *et al.* (2018) mentioned the empirical knowledge of family farmers as one of the factors that sustain decision making, with rules and institutions established by them, knowledge that often results from the exchange of information between them. The exchange of knowledge between farmers can be interpreted as a strategy in the face of agrarian social vulnerabilities. Gazolla; Schneider (2007) even highlighted “self-consumption” (part reserve of production for consumption by the farmers themselves) as an influential factor in the relations of material and immaterial exchanges between families.

The IDM is characterized by the variety of strategies used to control plant diseases. The social aspects, also understood by the IDM, are evidenced in these results especially when there are discussions about the executable strategies and the actions that are culturally incorporated by the farmers. In this case, there is emphasis on the practice of eliminating diseased plants, which presented significant results for “often or sometimes”. Although this practice does not come from formal knowledge, it is in common use among producers with a higher degree of technology. The practices of seed treatment and change of planting location are little or never carried out. This reflects the profile of a farmer who,

due to the difficulty of accessing technology, that is, due to cultural attributes, preserves greater traditionality.

Actions to manage crops tend to be different according to the scale of production, and according to the strategies applicable to farmers. Grasswitz (2019) studied the integrated pest management and argued that there are more viable strategies for small-scale farms than for larger production scales. A non-fixed schedule for applying fungicides is already a reality for most farmers, which may be in accordance with the management possibilities they have. Haye *et al.* (2016) highlighted the decrease in the interval between harvests as strategies to decrease insect infestations.

This complex conjuncture allows a greater understanding of the characteristics of rural activities and family farming in Ribeira Valley. This is what Fidelis; Bergamasco (2015) demonstrated when they realized that the quilombo family farmers in this region are connected to the notions of modernity in many social aspects, but maintain traditional practices, having also difficulties with the traditional model.

It is significant to know that most of the small producers in Vale do Ribeira incorporate disease control techniques. Extension and technical assistance can emphasize the importance of these practices in your day-to-day production. However, for this advice to be efficient, it is necessary that the considerations and knowledge of these farmers are considered and associated with scientific knowledge. Testen *et al.* (2018) affirm that farmers can better allocate limited resources for the management of diseases in tomatoes (*Solanum lycopersicum* L.) when they better understand the diseases by combining their own local knowledge. They also state that information from research on plant diseases and ethnophytopathological studies can be associated so that better tools are obtained for farmers to know and manage diseases in crops.

6 CONCLUSION

From the results presented, we conclude that the IDM, even though it is not perceived by family farmers as a concept, is a reality for them. The actions of IDM can, and should, continue to be part of the routine of the production process,

collaborating with the productive success and, therefore, with the socioeconomic reality of family farmers in Ribeira Valley, São Paulo State, Brazil.

IDM strategies can also have a lasting perspective, strengthening the traditional knowledge and technological advances bonds in the plant diseases management. With the IDM popularization, a social technology strengthening for advances in family farming is expected.

Our work also impacts other studies in other locations, especially those areas where family farming influences food security. It is also hoped that this study and its methodology will help other researchers to diagnose techniques popularly adopted in different cropping systems, which contributes to the dialogic exchange of knowledge between the scientific community and farmers.

The financial contribution of this study was a conditioning factor for the selection of the target audience interviewed. Other funding for future research can help increase the target audience, which allows for more assertive agricultural solutions.

ACKNOWLEDGMENTS

We thank the farmers involved in this study, the Integral Technical Assistance Coordination (CATI), a public coordination agency in São Paulo State, and the Pró-reitoria de Extensão Universitária e Cultura (PROEC) of Unesp for financial support.

REFERENCES

ALMEIDA, L. C. F. *et al.* Seriam as questões ambientais entraves ao desenvolvimento do Vale do Ribeira? **Cadernos CERU**, São Paulo, v. 28, n. 01, p. 93-104. Aug. 2017. Available at: <https://www.revistas.usp.br/ceru/article/view/137126>. Accessed on September 22, 2021.

ALMEIDA, J. A. **Pesquisa em extensão rural**: um manual de metodologia. 1st ed. Brasília: Associação Brasileira de Educação Agrícola Superior ABEAS, 1989.

AMORIM L, REZENDE JAM & BERGAMIN FILHO A. 2011. **Manual de fitopatologia**. 4th ed. São Paulo: Ceres, 2011.

ANGELO, J. A. *et al.* Previsões e estimativas das safras agrícolas do estado de São Paulo, intenção de plantio do ano agrícola 2016/17, e levantamento final ano agrícola 2015/16. **Análises e Indicadores do Agronegócio**, São Paulo, v. 11, n. 01, p. 1-11. Nov. 2016. Available at: <http://www.iea.sp.gov.br/out/artigosai2.php?codTipo=2&ano=2016>. Accessed on September 22, 2021.

BERGAMIN FILHO, A.; AMORIM, L. **Doenças de plantas tropicais**: epidemiologia e controle econômico. 1st ed. Piracicaba: Ceres, 1996.

BRAZIL: Brazilian decree 9,064. Brasília (DF): Presidency of the Republic, 2017. Available at: <https://www2.camara.leg.br/legin/fed/decret/2017/decreto-9064-31-maio-2017-785001-publicacaooriginal-152929-pe.html>. Accessed on October 25, 2020.

BRAZIL: Brazilian law 11,326. 2006. Brasília (DF): Presidency of the Republic. Available at: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/lei/l11326.htm. Accessed on October 25, 2020.

BUAINAIN, A. M.; ROMEIRO, A.R.; GUANZIROLI, C. Agricultura familiar e o novo mundo rural. **Sociologias**, Porto Alegre, v. 5, n. 10, p. 312-347. Dec. 2003. Available at: <https://www.scielo.br/j/soc/a/yWYK66v4CJXDqsmKtVH5bkD/?lang=pt&format=pdf>. Accessed on September 22, 2021.

CHIARAPPA L. Possibility of supervised plant disease control in pest management systems. **FAO Plant Protection Bulletin**, Rome, v. 22, n. 01, p. 65-68. Dec. 1974. Available at: <https://agris.fao.org/agris-search/search.do?recordID=US201302781380>. Accessed on September 22, 2021.

CRESWELL, J. W. Mapping the developing landscape of mixed methods research. *In*: **SAGE handbook of mixed methods in social & behavioral research**. 2nd ed. Thousand Oaks: SAGE Publications, 2010.

DAWSON, B.; TRAPP, R. G. **Bioestatística básica e clínica**. 3rd ed. Rio de Janeiro: McGraw Hill Interamericana do Brasil, 2003.

DIEGUES, A. C. O Vale do Ribeira e litoral de São Paulo: meio ambiente, história e população. **CENPEC**, São Paulo, v. 01, n. 01, p. 1-41. Mar. 2007. Available at: <https://nupaub.fflch.usp.br/sites/nupaub.fflch.usp.br/files/color/cenpec.pdf>. Accessed on September 22, 2021.

DUFUMIER, M. Systèmes de production et développement agricole dans le Tiers Monde. **Les cahiers de la recherche développement**. Paris, v. 06, n. 01, p. 31-38. Jan. 1985. Available at: <https://revues.cirad.fr/index.php/crd/article/view/36439/36132>. Accessed on September 22, 2021.

FAO. Report of the first session of the FAO Panel of Experts on Integrated Pest Control. **UN Food and Agriculture Organization**. Rome, v. 01, n. 06, p. 133-142. Sept. 18-22, 1967. Available at: <http://www.fao.org/pest-and-pesticide-management/guidelines-standards/faowho-joint-meeting-on-pesticide-management-jmpm/meetings/meeting-reports/en/>. Accessed on September 22, 2021.

FERNANDES, R. S.; SILVA, A. M.; FALCÃO, R. P. A pluriatividade como fator de geração de renda para a agricultura familiar de Córrego Primavera. **Extensão Rural**, Santa Maria, v. 25, n. 02, p. 52-72. Jun., 2018. Available at: <https://periodicos.ufsm.br/extensaorural/article/view/29969>. Accessed on September 22, 2021.

FIDELIS, L. M.; BERGAMASCO, S. M. P. P. A família, suas mudanças e a manutenção dos saberes tradicionais na agricultura familiar em quilombos do Vale do Ribeira paranaense. **Divers@!**, Matinhos, v. 08, n. 02, p. 59-72. Dec., 2015. Available at: <https://revistas.ufpr.br/diver/article/view/45042/27424>. Accessed on September 22, 2021.

FREITAS, M. F. L. **Agricultura familiar e políticas de redução da pobreza rural: uma análise do perfil dos agricultores participantes do Programa de Aquisição de Alimentos (PAA) no município de Tunas do Paraná, Vale do Ribeira-PR**. 1st ed. Curitiba: Acervo digital UFPR, 2015.

GAZOLLA, M.; SCHNEIDER, S. A produção da autonomia: os "papeis" do autoconsumo na reprodução social dos agricultores familiares. **Estudos sociedade e agricultura**, Rio de Janeiro, v. 15, n. 01, p. 89-122. Apr, 2007. Available at: <https://revistaesa.com/ojs/index.php/esa/article/view/283>. Accessed on September 22, 2021.

GOMES, T. T.; DE ABREU, L. S. Percepções de agricultores familiares e técnicos da região do Vale do Ribeira (SP) sobre o mercado institucional. **Extensão Rural**, Santa Maria, v. 26, n. 04, p. 51-68. Oct., 2019. Available at: <https://periodicos.ufsm.br/extensaorural/article/view/30773/pdf>. Accessed on September 22, 2021.

GRASSWITZ, T. R. Integrated Pest Management (IPM) for small-scale farms in developed economies: Challenges and opportunities. **Insects**, Lyndonville, v. 10,

n.06, p. 179. June, 2019. Available at: <https://www.mdpi.com/2075-4450/10/6/179>. Accessed on September 22, 2021.

HAYE, T. *et al.* Current SWD IPM tactics and their practical implementation in fruit crops across different regions around the world. **Journal of pest Science**, v. 89, n. 03, p. 643-651. Feb., 2016. Available at: <https://link.springer.com/article/10.1007/s10340-016-0737-8>. Accessed on September 22, 2021.

[IBGE] The Brazilian Institute of Geography and Statistics. 2016. Tamanho dos municípios. Brasília (DF): IBGE; [accessed 2016 Nov 14]. [tps://www2.ibge.gov.br/home/geociencias/cartografia/default_territ_area.shtm](https://www2.ibge.gov.br/home/geociencias/cartografia/default_territ_area.shtm)

[IBGE] The Brazilian Institute of Geography and Statistics. 2017. Censo Agro. Brasília (DF): IBGE; [accessed 2020 Oct 25]. https://censos.ibge.gov.br/agro/2017/templates/censo_agro/resultadosagro/produtores.html

[IBGE] The Brazilian Institute of Geography and Statistics. 2020. Produção agrícola municipal. Brasília (DF): IBGE; [accessed 2020 Oct 25]. <https://sidra.ibge.gov.br/tabela/1618>

[IBGE] The Brazilian Institute of Geography and Statistics. 2024. Produção agrícola municipal. Brasília (DF): IBGE; [accessed 2024 Feb 08]. <https://sidra.ibge.gov.br/tabela/1618>

[IEA] The Institute of Agricultural Economics. 2020. Estatísticas da Produção Paulista. São Paulo (SP): IEA; [accessed 2020 Oct 25]. http://ciagri.iea.sp.gov.br/nia1/subjetiva.aspx?cod_sis=1&idioma=1

KHOURY, W. E.; MAKKOUK, K. Integrated plant disease management in developing countries. **Journal of Plant Pathology**, Pisa, v. 92, n. 04, p. 35-42. Dec., 2010. Available at: <http://www.sipav.org/main/jpp/index.php/jpp/article/view/340>. Accessed on September 22, 2021.

LARSON, R.; FARBER, B. **Estatística aplicada**. 4th ed. São Paulo: Pearson Prentice Hall, 2010.

LIKERT, R. A technique for the measurement of attitudes. **Archives of Psychology**, Bethel Park, v. 140, n. 01, p. 44-53. Dec., 1932. Available at: <https://psycnet.apa.org/record/1933-01885-001>. Accessed on September 22, 2021.

MARSDEN, T.; MUNTUN, R.; WARD, N. Incorporating social trajectories into uneven agrarian development: Farm businesses in upland and lowland Britain. **Sociologia Ruralis**, Trondheim, v. 32, n. 04, p. 408-430. Jan., 1992. Available at: <https://core.ac.uk/display/2773530>. Accessed on September 22, 2021.

MILONE, G. **Estatística geral e aplicada**. 2nd ed. São Paulo: Pioneira Thomson Learning, 2004.

MONTEIRO, J. E. B. A. **Agrometeorologia dos cultivos**: o fator meteorológico na produção agrícola. 1st ed. Brasília: INMET, 2009.

PEAT, J.; BARTON, B. **Medical Statistics**: a guide to data analysis and critical appraisal. 1st ed. Massachusetts: Blackwell Publishing, 2005.

ROMÃO, D. A. **Vale do Ribeira**: um ensaio para o desenvolvimento das comunidades rurais. 1st ed. Brasília: NEAD, 2006.

SHEKIN, D. **Handbook of parametric and nonparametric statistical procedures**. 3rd ed. New York: Chapman and Hall/CRC, 2003.

SCHNEIDER, S. Teoria social, agricultura familiar e pluriatividade. **Revista brasileira de ciências sociais**, v. 18, n. 51, p. 99-122. Feb., 2003. Available at: <https://www.scielo.br/j/rbcsoc/a/rztr5GB6thSx7TVPkw4wf7z/abstract/?lang=pt>. Accessed on September 22, 2021.

STERN, V. M. *et al.* The integration of chemical and biological control of the spotted alfalfa aphid: The integrated control concept. **Hilgardia**, Berkeley, v. 29, n. 02, p. 81-101. Oct., 1959. Available at: <http://hilgardia.ucanr.edu/Abstract/?a=hilg.v29n02p081>. Accessed on September 22, 2021.

TESTEN, A. L. *et al.*, Integrating ethnophytopathological knowledge and field surveys to improve tomato disease management in Tanzania. **Canadian Journal of Plant Pathology**, Winnipeg, v. 40, n. 01, p. 22-33. Feb., 2018. Available at: <https://www.tandfonline.com/doi/abs/10.1080/07060661.2017.1414881?journalCode=tcjp20>. Accessed on September 22, 2021.

ZADOKS, J. C.; SCHEIN, R. D. **Epidemiology and plant disease management**. 1st ed. New York: Oxford University Press, 1979.

WÓJCIK, M.; JEZIORSKA-BIEL, P.; CZAPIEWSKI, K. Between words: A generational discussion about farming knowledge sources. **Journal of Rural Studies**, Oxford, v. 64, n. 01, p. 130-141. Apr., 2019. Available at: https://www.sciencedirect.com/science/article/pii/S0743016718306260?casa_token=dOpBdVrUepQAAAAA:l5cOFcKF-SxLLNGKCr-gRAcl8U_z1BwbzMdpXPActO6R4oyC_9RSfNiX046zyeKKAUPFplSf1A. Accessed on September 22, 2021.

ZUIN, L. F. S.; ZUIN, P. B.; DÍAZ MANRIQUE, M. A. A comunicação dialógica como fator determinante para os processos de ensino-aprendizagem que ocorrem na capacitação rural: um estudo de caso em um órgão público de extensão localizado no interior do Estado de São Paulo. **Ciência Rural**, Santa Maria, v. 41, n. 05, p. 917-923. 2011. Available at: <https://pesquisa.bvsalud.org/portal/resource/pt/vti-707529>. Accessed on September 22, 2021.